Methods Map

STAT 109: Introductory Biostatistics

Methods Table

This page is our statistics "toolbox." It is organized by:

- what we want to do (descriptive: plot/number summary, or inferential: test/CI/prediction), and
- what we have (the number and type of variables).

For the table below:

- columns = what we have (number/type of variables),
- rows = what we want to do (describe, infer a parameter, infer an individual value).

One toolbox table

What we want	One categorical	One numerical	Two categorical (2+ categories allowed)	EV categorical + RV numerical (2 EV groups)	EV categorical + RV numerical (3+ EV groups)	Two numerical	RV binary + EV predictor(s)
Describe: number summary	Counts, proportions, percentages (`count`, `mutate`)	Mean/median; SD/range/IQR (`summarize`)	Two-way counts + row/column/grand proportions	Group-wise means/SDs by category (`group_by`)	Group-wise means/SDs by category (`group_by`)	Correlation `cor(x,y)` + slope/intercept summaries	Class counts and proportions by predictor groups
Describe: plot	Bar plot (`geom_bar` or `geom_col`)	Histogram / boxplot / dotplot	Stacked or 100% stacked bar (`geom_col(position = "fill")`)	Boxplot/histogram/dotplot per group	Boxplot per group (ANOVA context)	Scatterplot (`geom_point`)	Bar plots by class or probability plot vs predictor
Infer parameter: notation	p	μ	Association (χ^2 independence); 2×2: $p_1 - p_2$	$\mu_1 - \mu_2$ (numerical RV) or $p_1 - p_2$ (binary RV)	$\mu_1, \dots, \mu_k$ (ANOVA across EV groups)	β_0, β_1	Logistic β; prob. RV given EV
Infer parameter: test name + sample null	One-proportion test, $H_0 : p = p_0$	One-sample (or paired) t-test, $H_0 : \mu = \mu_0$	Chi-square independence, H_0: EV and RV independent	Two-sample t-test $H_0 : \mu_1 - \mu_2 = 0$, or two-proportion $H_0 : p_1 - p_2 = 0$	One-way ANOVA $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$	Slope test $H_0 : \beta_1 = 0$	Logistic slope $H_0 : \beta_1 = 0$ (or relevant coefficient $= 0$)

What we want	One categorical	One numerical	Two categorical (2+ categories allowed)	EV categorical + RV numerical (2 EV groups)	EV categorical + RV numerical (3+ EV groups)	Two numerical	RV binary + EV predictor(s)
Infer parameter: test R code	`binom.test()` (exact) or `prop.test()`	`t.test(x)` or paired `t.test(x,y,paired=TRUE)`	`chisq.test(tab)`	`t.test(y ~ group, var.equal = FALSE)` or `prop.test(...)`	`fit <- lm(y ~ group); anova(fit)`	`fit <- lm(y ~ x); summary(fit)`	`fit <- glm(y ~ x_or_group, family = binomial); summary(fit)`
Infer parameter: test conditions	Binomial / SRS; for z approx need $np \geq 10$ and $n(1-p) \geq 10$	Random sample (or paired) + large n or roughly normal population/differences	Independent obs.; expected counts typically ≥ 5	Two samples or random assignment; conditions for chosen method	Independent obs.; roughly normal within groups; similar spread	Linear form; independent obs.; residual assumptions for t-based inference	Binary RV; independent obs.; appropriate model form
Infer parameter: CI name	One-proportion CI	One-sample (or paired) t-interval	Context-dependent (cell/row/column or follow-up)	Two-sample t-interval or two-proportion CI	Post-hoc / group-mean intervals after ANOVA	CI for β_1; CI for mean response at x_0	CI for logistic β
Infer parameter: CI R code	`prop.test()`	`t.test()`	(Context-dependent; often not a single omnibus CI)	`t.test(...)` or `prop.test(...)`	`fit <- lm(...);` then follow-up contrasts/intervals as needed	`confint(fit, "x"); predict(fit, newdata=..., interval="confidence")`	`confint(fit, "x_or_group")`
Infer parameter: CI conditions	Same as proportion-test approximation	Same as t-test conditions	Depends on interval used; independence remains key	Same as chosen two-group method	Same assumptions as ANOVA model	Same regression assumptions as above	Same logistic-model assumptions as above
Infer individual value: notation	Individual category outcome (usually not main target)	Individual future RV	Individual future cell/category (usually not main target)	Individual future RV in a selected EV group	Individual future RV in selected EV group level	Individual future RV at x_0	Individual $P(\text{RV} = 1 \mid \text{EV})$ and class after cutoff
Infer individual value: method name	(Usually classify/report probability)	Prediction interval for one value	(Usually classify/report probability)	EV-group-based prediction interval for one value	EV-group-based prediction interval for one value	Linear-model prediction interval	Logistic probability prediction + cutoff classification

What we want	One categorical	One numerical	Two categorical (2+ categories allowed)	EV categorical + RV numerical (2 EV groups)	EV categorical + RV numerical (3+ EV groups)	Two numerical	RV binary + EV predictor(s)
Infer individual value: R code	`prop.test`/model-based probability methods	`predict(fit, newdata=..., interval="prediction")`	`chisq` context not usually individual prediction	`fit <- lm(y ~ group); predict(fit, newdata=..., interval="prediction")`	`fit <- lm(y ~ group); predict(fit, newdata=..., interval="prediction")`	`fit <- lm(y ~ x); predict(fit, newdata=..., interval="prediction")`	`fit <- glm(..., family=binomial); predict(fit, newdata=..., type="response")`
Infer individual value: conditions	Use caution; not usually central target	Linear-model assumptions; interval is for **individual** not mean	Not usually the main inferential target	Independent observations; prediction interval for individual not mean	Same as linear-model assumptions across groups	Linear form + residual diagnostics; independent observations	Binary outcome; independent observations; choose and justify cutoff

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Notes

- **EV** = explanatory variable (sometimes called treatment/predictor, depending on context).
- **RV** = response variable (aka outcome variable).
- For any inferential row, always check:
  1. sample size(s),
  2. sampling method (random sample or independent/binomial trials),
  3. study design (observational vs experimental, paired vs independent),
  4. model assumptions where relevant.